

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only OR for Specialist Use only

Erlocip 25 / 100 / 150 Film Coated Tablet

Erlotinib Hydrochloride 25 / 100 / 150

COMPOSITION

Erlocip 25

Each film coated tablet contains

Erlotinib hydrochloride equivalent to Erlotinib.....25 mg

Colour: Titanium dioxide

Erlocip 100

Each film coated tablet contains

Erlotinib hydrochloride equivalent to Erlotinib.....100 mg

Colour: Titanium dioxide

Erlocip 150

Each film coated tablet contains

Erlotinib hydrochloride equivalent to Erlotinib.....150 mg

Colour: Titanium dioxide

DOSAGE FORM

Film Coated Tablet

PRODUCT DESCRIPTION

Erlocip 25

White, round, biconvex, film coated tablet debossed with "C1" on one side and plain on the other side.

Erlocip 100

White, round, biconvex, film coated tablet debossed with "C2" on one side and plain on the other side.

Erlocip 150

White, round, biconvex, film coated tablet debossed with 'C3' on one side and plain on other side.

PHARMACOLOGY

Pharmacodynamics

Mechanism of action

Erlotinib potently inhibits the intracellular phosphorylation of HER1/EGFR receptor. HER1/EGFR receptor is expressed on the cell surface of normal cells and cancer cells. In non-clinical models, inhibition of EGFR phosphotyrosine results in cell stasis and/or death.

Pharmacokinetics

Exposure

Following a 150 mg oral dose of erlotinib, at steady state, the median time to reach maximum plasma concentrations is approximately 4 hours with median maximum plasma concentrations achieved of 1,995 ng/ml. Prior to the next dose at 24 hours, the median minimum plasma concentrations are 1,238 ng/ml. Median AUC achieved during the dosing interval at steady state are 41.300 mcg*hr/ml.

Absorption

Oral erlotinib is well absorbed and has an extended absorption phase, with mean peak plasma levels occurring at 4 hours after oral dosing, provided an estimate of bioavailability of 59%. The exposure after an oral dose may be increased by food.

Following absorption, erlotinib is highly bound in blood, with approximately 95% bound to blood components, primarily to plasma proteins (i.e. albumin and alpha-1 acid glycoprotein [AAG]), with a free fraction of approximately 5%.

Distribution

Erlotinib has a mean apparent volume of distribution of 232 L and distributes into tumor tissue of humans.

Metabolism

Erlotinib is metabolised in humans by hepatic cytochrome P450 enzymes, primarily by CYP3A4 and to a lesser extent by CYP1A2. Extrahepatic metabolism by CYP3A4 in intestine, CYP1A1 in lung, and CYP1B1 in tumour tissue potentially contribute to the metabolic clearance of erlotinib. There are three main metabolic pathways identified: 1) O- demethylation of either side chain or both, followed by oxidation to the carboxylic acids; 2) oxidation of the acetylene moiety followed by hydrolysis to the aryl carboxylic acid; and 3) aromatic hydroxylation of the phenyl-acetylene moiety. The primary metabolites of erlotinib produced by O-demethylation of either side chain have comparable potency to erlotinib in preclinical *in-vitro* assays *in vivo* tumor models. They are present in plasma at levels that are <10% of erlotinib and display similar pharmacokinetics as erlotinib.

Elimination

The metabolites and trace amounts of erlotinib are excreted predominantly via the feces (>90%), with renal elimination accounting for only a small amount of an oral dose.

Clearance

The time to reach steady state plasma concentration would be expected to occur in approximately 7-8 days. No significant relationships between predicted apparent clearance and patient age, body weight, gender, and ethnicity were observed.

Patient factors, which correlate with erlotinib pharmacokinetics, are serum total bilirubin, AAG concentrations and current smoking. Increased serum concentrations of total bilirubin and AAG concentrations were associated with a slower rate of erlotinib clearance. Smokers had a higher rate of erlotinib clearance (see *Drug Interactions*).

Pharmacokinetics in Special Populations

There have been no specific studies in pediatric or elderly patients.

Hepatic impairment

Erlotinib is mainly cleared by liver. Erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function including patients with primary liver cancer or hepatic metastases.

Renal impairment

Erlotinib and its metabolites are not significantly excreted by the kidneys, as less than 9% of a single dose is excreted in the urine. No clinical studies have been conducted in patients with compromised renal function.

Smokers

Cigarette smoking leads to increased clearance of, and decreased exposure to erlotinib. This reduced exposure in current smokers is presumably due to induction of CYP1A1 in lung and CYP1A2 in the liver.

INDICATIONS

Non-small Cell Lung Cancer

Erlotinib is indicated for the first-line treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with EGFR activating mutations.

Erlotinib is indicated for maintenance treatment in patients with locally advanced or metastatic NSCLC with EGFR activating mutations who have not progressed after first-line chemotherapy. Erlotinib is also indicated for the treatment of patients with locally advanced or metastatic NSCLC after failure of at least one prior chemotherapy regimen.

Pancreatic Cancer

Erlotinib in combination with gemcitabine is indicated for the treatment of patients with locally advanced, unresectable or metastatic pancreatic cancer.

DOSAGE AND METHOD OF ADMINISTRATION

Posology

Non-small cell lung cancer

EGFR mutation testing should be performed prior to initiation of erlotinib therapy as first-line or maintenance therapy in patients with locally advanced or metastatic NSCLC.

The recommended daily dose of erlotinib is 150 mg taken at least one hour before or two hours after the ingestion of food.

Pancreatic cancer

The recommended daily dose of erlotinib is 100 mg taken at least one hour before or two hours after the ingestion of food, in combination with gemcitabine (see the package insert of gemcitabine for the pancreatic cancer indication)

Special dosage instructions

Drug Interactions

Concomitant use of CYP 3A4 substrates and modulators may require dose adjustment (see *Drug Interactions*).

When dose adjustment is necessary, it is recommended to reduce in 50 mg steps (see *Warnings and Precautions* and *Drug Interactions*)

Hepatic impairment

Erlotinib is eliminated by hepatic metabolism and biliary excretion. Although erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function, caution should be used when administering erlotinib to patients with hepatic impairment. Dose reduction or interruption of erlotinib should be considered if severe adverse reactions occur. Safety and efficacy have not been studied in patients with severe hepatic impairment (see *Warnings and Precautions* and *Pharmacokinetics*).

Renal impairment

The safety and efficacy of erlotinib has not been studied in patients with renal impairment. (see *Pharmacokinetics*).

Paediatric Use

The safety and efficacy of erlotinib has not been studied in patients under the age of 18 years.

Smokers

Cigarette smoking has been shown to reduce erlotinib exposure by 50-60%. The maximum tolerated dose of erlotinib in NSCLC patients who currently smoke cigarettes was 300 mg. The 300 mg dose did not show improved efficacy in second line treatment after failure of chemotherapy compared to the recommended 150 mg dose in patients who continue to smoke cigarettes (see *Drug Interactions and Pharmacokinetics*).

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Erlocip is contraindicated in patients with severe hypersensitivity to erlotinib or to any component of Erlocip.

WARNINGS AND PRECAUTIONS

Interstitial Lung Disease: Cases of interstitial lung disease (ILD)-like events, including fatalities were uncommon in patients receiving erlotinib for treatment of NSCLC (non-small cell lung carcinoma), pancreatic cancer or other advanced solid tumors. Some examples of reported diagnoses in patients suspected of having ILD-like events include pneumonitis, radiation pneumonitis, hypersensitivity pneumonitis, interstitial pneumonia, interstitial lung disease, obliterative bronchiolitis, pulmonary fibrosis, Acute Respiratory Distress Syndrome (ARDS), alveolitis, and lung infiltration. Symptoms started from a few days to several months after initiating erlotinib therapy. Confounding or contributing factors such as concomitant or prior chemotherapy, prior radiotherapy, pre-existing parenchymal lung disease, metastatic lung disease, or pulmonary infections were frequent.

In patients who develop acute onset of new and/or progressive unexplained pulmonary symptoms, such as dyspnea, cough and fever, erlotinib therapy should be interrupted pending diagnostic evaluation. If ILD is diagnosed, erlotinib should be discontinued and appropriate treatment initiated as necessary (see *Undesirable effects*).

Diarrhea, Dehydration, Electrolyte imbalance and Renal failure: Diarrhea has occurred in patients on erlotinib and moderate or severe diarrhea should be treated with loperamide. In some cases, dose reduction may be necessary. In the event of severe or persistent diarrhea, nausea, anorexia, or vomiting associated with dehydration, erlotinib therapy should be interrupted

and appropriate measures should be taken to treat the dehydration (see *Undesirable effects*). There have been rare reports of hypokalemia and renal failure (including fatalities). Some reports of renal failure were secondary to severe dehydration due to diarrhea, vomiting and/or anorexia while others were confounded by concomitant chemotherapy. In more severe or persistent cases of diarrhea, or cases leading to dehydration, particularly in groups of patients with aggravating risk factors (concomitant medications, symptoms or diseases or other predisposing conditions including advanced age), erlotinib therapy should be interrupted and appropriate measures should be taken to intensively rehydrate the patients intravenously. In addition, renal function and serum electrolytes including potassium should be monitored in patient at risk of dehydration.

Hepatitis, hepatic failure: Rare cases of hepatic failure (including fatalities) have been reported during use of erlotinib. Confounding factors have included pre-existing liver disease or concomitant hepatotoxic medications. Therefore, in such patients, periodic liver function testing should be considered. Erlotinib dosing should be interrupted if changes in liver function are severe (see *Undesirable effects*).

Hepatic impairment

Erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function including patients with primary liver cancer or hepatic metastases. Safety and efficacy have not been studied in patients with severe hepatic impairment.

Gastrointestinal Perforation: Patients receiving erlotinib are at increased risk of developing gastrointestinal perforation, which was observed uncommonly (including some cases with failure outcome). Patients receiving concomitant anti-angiogenic agents, corticosteroids, NSAIDs, and/or taxane based chemotherapy, or who have prior history of peptic ulceration or diverticular disease are at increased risk. Erlotinib should be permanently discontinued in patients who develop gastrointestinal perforation (see *Undesirable effects*).

Bullous and exfoliative skin disorders: Bullous, blistering and exfoliative skin conditions have been reported, including very rare cases suggestive of Stevens-Johnson syndrome/Toxic epidermal necrolysis, which in some cases were fatal (see *Undesirable effects*). Erlotinib treatment should be interrupted or discontinued if the patient develops severe bullous, blistering or exfoliating conditions.

Ocular Disorders: Very rare cases of corneal perforation or ulceration have been reported during use of erlotinib. Other ocular disorders including abnormal eyelash growth, keratoconjunctivitis sicca or keratitis have been observed with erlotinib treatment which are also risk factors for corneal perforation/ulceration. Erlotinib therapy should be interrupted or discontinued if patients present with acute/worsening ocular disorders such as eye pain (see *Undesirable effects*).

Interactions: Erlotinib has a potential for clinically significant drug-drug interactions (see *Drug Interactions*).

Lactose intolerance: Erlotinib tablets contain lactose. Patients with rare hereditary problems of fructose intolerance, galactose intolerance, galactosaemia or glucose-galactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Erlotinib is metabolized in the liver by the hepatic cytochromes in humans, primarily CYP3A4 and to a lesser extent by CYP2A6, and the pulmonary isoform CYP1A1. Potential interactions may occur with drugs which are metabolized by, or are inhibitors or inducers of, these enzymes.

Potent inhibitors of CYP3A4 activity decrease erlotinib metabolism and increase erlotinib plasma concentrations. Inhibition of CYP3A4 metabolism by ketoconazole (200 mg po BID for 5 days) resulted in increased exposure to erlotinib. When erlotinib was co-administered with ciprofloxacin, an inhibitor of both CYP3A4 and CYP1A2, the erlotinib exposure [AUC] and maximum concentration (C_{max}) increased by 39% and 17%, respectively. Therefore caution should be used when administering erlotinib, with potent CYP3A4 or combined CYP3A4/CYP1A2 inhibitors. In these situations, the dose of erlotinib should be reduced if toxicity is observed.

Potent inducers of CYP3A4 activity, example, rifampicin, increase erlotinib metabolism and significantly decrease erlotinib plasma concentrations.

For patients who required concomitant treatment with erlotinib and a potent CYP3A4 inducer such as rifampicin an increase in dose to 300 mg should be considered while their safety (see *Warnings and Precautions*) is closely monitored, and if well tolerated for more than 2 weeks, further increase to 450 mg could be considered with close safety monitoring. Higher doses have not been studied in this setting.

Pre-treatment or co-administration of erlotinib did not alter the clearance of the prototypical CYP3A4 substrates midazolam and erythromycin. Significant interactions with the clearance of other CYP3A4 substrates are therefore unlikely. Oral availability of midazolam did appear to decrease by up to 24 %, which was however not attributed to effects on CYP3A4 activity.

The solubility of erlotinib is pH dependent. Erlotinib solubility decreases as pH increases. Drugs that alter the pH of the upper GI tract may alter the solubility of erlotinib and hence its bioavailability. Co-administration of drugs reducing gastric acid production, such as omeprazole and ranitidine, with erlotinib should be avoided where possible. Increasing the dose of erlotinib when co-administered with such agents is not likely to compensate for this loss of exposure. If patients need to be treated with such drugs, then a H₂-receptor antagonist such as ranitidine should be considered and used, in a staggered manner. Erlotinib must be taken at least 2 hours before or 10 hours after the H₂-receptor antagonist dosing.

Interaction with coumarin-derived anticoagulants, including warfarin, leading to increased International Normalized Ratio (INR) and bleeding events, which in some cases were fatal. Patients taking coumarin-derived anticoagulants should be monitored regularly for any changes in prothrombin time or INR.

The combination of erlotinib and a statin may increase the potential for statin-induced myopathy, including rhabdomyolysis, which was observed rarely.

Smokers should be advised to stop smoking as cigarettes smoking, which is known to induce CYP1A1 and CYP1A2, has been shown to reduce erlotinib exposure by 50-60% (see *Dosage and Method of Administration* and *Pharmacokinetics*).

There were no significant effects of gemcitabine on the pharmacokinetics of erlotinib nor were there significant effects of erlotinib on the pharmacokinetics of gemcitabine.

FERTILITY, PREGNANCY AND LACTATION

Pregnancy

There are no adequate or well controlled studies in pregnant women using erlotinib.

Fertility

The potential risk for humans is unknown. Women of childbearing potential must be advised to avoid pregnancy while on erlotinib. Adequate contraceptive methods should be used during therapy, and for at least 2 weeks after completing therapy. Treatment should only be continued in pregnant women if the potential benefit to the mother outweighs the risk to the foetus.

Lactation

It is not known whether erlotinib is excreted in human milk. As the potential for harm to the nursing infant is unknown, mothers should be advised against breastfeeding while receiving erlotinib and for at least 2 weeks after the final dose.

UNDESIRABLE EFFECTS

The following terms are used to rank the undesirable effects by frequency: very common; common; uncommon; rare; very rare including isolated reports.

System Organ Class	Adverse Events Description	Frequency
Infections and infestations	Infection	Very common
Metabolism and nutrition disorders	Anorexia, weight decreased	Very common
Eye disorders	Conjunctivitis, keratoconjunctivitis sicca Keratitis, conjunctivitis Corneal ulcerations or perforations Abnormal eyelash growth (in-growing eyelash, excessive growth and thickening of the eyelashes), uveitis	Very common Common Very rare Unknown
Psychiatric disorders	Depression	Very common
Nervous system disorders	Headache, neuropathy	Very common
Respiratory, thoracic and mediastinal disorders	Dyspnea, cough Epistaxis Serious interstitial lung disease (ILD)-like events	Very common Common Uncommon
Gastrointestinal disorders	Diarrhea, nausea, vomiting, stomatitis, abdominal pain, dyspepsia, flatulence Gastrointestinal bleeding Gastrointestinal perforations	Very common Common Uncommon
Skin and subcutaneous tissue disorders	Rash, pruritus, dry skin, alopecia Acne, dermatitis acneiform and folliculitis, skin fissures associated with rash and dry skin, Hyperpigmentation, hair and nail changes, mostly non-serious, e.g. hirsutism, eyelash / eyebrow changes, paronychia, brittle and loose nails Bullous, blistering and exfoliative skin conditions, Stevens-Johnson syndrome / Toxic epidermal necrolysis	Very common Common Uncommon Very rare

General disorders & administration site conditions	Fatigue, pyrexia, rigors	Very common
Hepatobiliary disorders	Liver function test abnormalities (raised ALT, AST, bilirubin) Hepatic failure (including fatalities)	Common Rare

Skin and subcutaneous tissue disorders:

Rash has been reported very commonly in patients receiving erlotinib and in general, manifests as a mild or moderate erythematous and papulopustular rash, which may occur or worsen in sun exposed areas. For patients who are exposed to sun, protective clothing, and/or use of sunscreen (e.g. mineral-containing) may be advisable.

Reporting of Suspected Adverse Reaction

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the local reporting system.

OVERDOSE

Severe adverse events such as diarrhea, rash, and possibly liver transaminase elevation may occur above the recommended dose of 150 mg. In case of suspected overdose erlotinib should be withheld and symptomatic treatment administered.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Erlotinib has no or negligible influence on the ability to drive and use machines.

INCOMPATIBILITIES

Not applicable.

STORAGE AND HANDLING INSTRUCTIONS

Store below 30°C.

SHELF LIFE

24 months

PACKAGING INFORMATION

1 blister pack of 10 tablets composed of clear PVC film / Plain aluminum blister foil.

PRODUCT REGISTRATION HOLDER

CIPLA MALAYSIA SDN BHD

Suite 1101, Amcorp Tower,

Amcorp Trade Centre

18, Persiaran Barat, 46050

Petaling Jaya, Selangor, Malaysia

MANUFACTURER

CIPLA LTD.

S-103 to S-105, S-107 to S-112, L-138,

L-147, L-147/1 to L-143/3 & L147/A

Verna Industrial Estate,

Verna Goa India

DATE OF REVISION

August 2022